

Unifying Special Relativity and Quantum Mechanics

Lecture 29 — The two-derivative CCWZ Lagrangian, and the Higgs Mechanism (abelian case)

(the unique two-derivative invariant $\text{tr}(d_\mu d^\mu)$, the leading EFT Lagrangian and its cutoff f_π , the validity of the derivative expansion, final considerations on Goldstone bosons; then: the abelian Higgs model, the unitary gauge, the Proca mass $m_A^2 = g^2 v^2$, the “eaten” Goldstone, and longitudinal gauge-boson scattering)

Typed version of the handwritten notes, with added explanations in blue boxes

Added explanation (not in the original notes) — How to read this document

Conventions. Exactly as in Lectures 1–28: normal text and the yellow / pink / grey boxes are a faithful transcription of the handwritten sheets, and the text colours follow the pens of the original (blue, magenta, green, red, violet, pencil, brown). Everything inside a blue box like this one is material I have added; it is not in the original notes.

Pagination — read this carefully, there are three things going on.

1. **The first two sheets are a repeat.** This scan opens at sheet 9 of the CCWZ chapter, but Lecture 28 already ran [p. 1*****]–[p. 10*****]. Sheets 9 and 10 are therefore duplicated from the end of that scan. Section 1 records their content *in full* — they are short, and they carry the single most important formula of the whole construction — clearly marked as a repeat.
2. **The CCWZ chapter then ends.** Sheets 11–13, [p. 11*****]–[p. 13*****], are new: they close the chapter with the EFT Lagrangian, its cutoff, and a page of “final considerations” that summarises the whole Goldstone story of Lectures 24–29.
3. **A new chapter begins mid-scan, after a blank page.** PDF page 6 is blank (the back of a sheet). Then the numbering restarts at 1 for a new chapter, *The Higgs mechanism*. That is the *fourteenth* restart of the notes, so those sheets are marked with *fourteen* stars: [p. 1*****]–[p. 8*****]. Thirteen stars means the CCWZ chapter; fourteen stars means the Higgs chapter. The two star-counts are not a typo — count them if in doubt.

So: PDF page n is CCWZ sheet $n + 8$ for $n \leq 5$; page 6 is blank; and PDF page n is Higgs sheet $n - 6$ for $n \geq 7$. Thirteen sheets of content, nothing missing, nothing out of order.

It stops mid-development, as usual. The last sheet, [p. 8*****], sets up the tree-level scattering of *longitudinal* gauge bosons, writes the full $s+t+u$ amplitude, draws the energy-scale ladder, and breaks off at the words “we use the UR limit for the gauge boson”. Section 10 carries the calculation to its punchline — which is the point of the whole exercise — and flags it clearly as added.

Slips and hard-to-read spots flagged in place.

1. On [p. 10*****] the four-derivative invariant is written $(\nabla_\mu d_\nu^\alpha)(\nabla^\mu d^{\alpha,\mu})$; the repeated μ is a slip for ν , i.e. $(\nabla_\mu d_\nu^\alpha)(\nabla^\mu d^{\alpha,\nu})$. (Same slip as in Lecture 28, same sheet.)
2. On [p. 10*****] the dummy contraction labels drift ($f_{\rho\gamma\alpha}$ versus $f_{\beta\gamma\alpha}$ between the two brackets); the names of dummy indices are irrelevant and the result is unaffected.
3. On [p. 11*****] the power of f_π inside the $\mathcal{O}(\cdot)$ symbol is hard to read in the scan. On dimensional grounds it must be $\mathcal{O}((\partial\pi)^2\pi^2/f_\pi^2)$ — and this is exactly what [p. 12*****] then uses. Written that way here.
4. On [p. 2*****] the shifted parametrisation is written $e^{i\pi(x)/v}$ in one place; throughout the abelian chapter the Goldstone decay constant *is* v , so this is consistent, not a slip — but note the change of notation from f_π of the previous chapter to v here.

5. On [p. 5*****] the Proca mass term is written $\frac{1}{2}g^2v^2A_\mu A^\mu$ and the mass is boxed as $m_A^2 = g^2v^2$. Consistent (see the blue box there for the factor-of-2 bookkeeping).

Notation carried over. From the CCWZ chapter: T^a unbroken ($a = 1, \dots, \dim H$), X^α broken ($\alpha = 1, \dots, \dim G - \dim H$), $U(\pi) = \exp(i\pi^\alpha(x)X^\alpha/f_\pi)$ the Goldstone matrix, $U^{-1}\partial_\mu U = id_\mu + iE_\mu$ the Maurer–Cartan form with vielbein d_μ^α and connection E_μ^a , $(t_\pi^a)_{\alpha\beta} = -if_{a\alpha\beta}$ the representation R_π of H carried by the Goldstones. From the new chapter: Φ a complex scalar of charge 1, g the gauge coupling, $v = \sqrt{\mu^2/\lambda}$, h the Higgs, π the Goldstone, and the subscript “ u ” meaning “in the unitary gauge”.

Added explanation (not in the original notes) — Where we left off — recap of Lecture 28

Lecture 28 built the CCWZ machinery. The essential chain was:

- Every group element factorises as $g = \sigma(z)h$: a *coset representative* built from broken generators, times an element of H . The Goldstone matrix $U(\pi) = \exp(i\pi^\alpha X^\alpha/f_\pi)$ is such a representative, and the π^α are coordinates on the tangent plane to the vacuum manifold $\mathcal{M}_0 \cong G/H$.
- Acting with $g \in G$ and refactorising gives the master rule $gU(\pi) = U(\pi^g)h(\pi, g)$, i.e. $U(\pi) \xrightarrow{G} U(\pi^g) = gU(\pi)h(\pi, g)^{-1}$, with h depending on the *fields* as well as on g — this field dependence is the technical content of “non-linear realisation”.
- The object $U^{-1}\partial_\mu U$ (the Maurer–Cartan form) splits uniquely on the algebra, $U^{-1}\partial_\mu U = id_\mu^\alpha X^\alpha + iE_\mu^a T^a$, with

$$d_\mu^\alpha = \frac{\partial_\mu \pi^\alpha}{f_\pi} - \frac{(\partial_\mu \pi^\beta)\pi^\gamma f_{\beta\gamma\alpha}}{2f_\pi^2} + \dots, \quad E_\mu^a = -\frac{f_{\alpha\beta a}}{2f_\pi^2}(\partial_\mu \pi^\alpha)\pi^\beta + \dots$$

- Under the *full* G : $d_\mu \rightarrow h d_\mu h^{-1}$ (homogeneous!) and $E_\mu \rightarrow h[E_\mu - i\partial_\mu]h^{-1}$ (inhomogeneous — exactly a gauge field of a local H). So d_μ^α transforms under *all* of G the way the Goldstones transform under H alone; E_μ^a may only appear through covariant derivatives and field strengths.

The scan of Lecture 28 stopped in the middle of the sheet that begins “note that, in terms of derivatives...”. **That is exactly where this scan opens** — which is why the first two sheets here are a repeat. From [p. 11*****] onwards everything is new.

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PART I — Closing the CCWZ chapter [p. 9***]–[p. 13*****]**
1 The duplicated sheets [p. 9***]–[p. 10*****]**
Added explanation (not in the original notes) — These two sheets already appeared at the end of Lecture 28

Sheets [p. 9*****] and [p. 10*****] are physically the same pages as the last two of the Lecture 28 scan, re-scanned here. Unlike the duplication in Lecture 28 (which I compressed), I transcribe these two *in full*: they contain the single formula the entire CCWZ construction was built to produce, and Sections 2ff. refer back to them line by line. For the surrounding discussion see Lecture 28, Section 9.

Note that, in terms of derivatives,

$$[d_\mu^\alpha] = [\partial], \quad [E_\mu^a] = [\partial]$$

i) We write an invariant using d_μ^α as follows

$$\text{tr}(d_\mu d_\nu) \xrightarrow{G} \text{tr}(h d_\mu h^{-1} h d_\nu h^{-1}) = \text{tr}(h d_\mu d_\nu h^{-1}) = \text{tr}(d_\mu d_\nu), \quad \text{invariant}$$

Note that we can also write

$$\text{tr}(d_\mu d_\nu) = \text{tr}(d_\mu^\alpha X^\alpha d_\nu^\beta X^\beta) = d_\mu^\alpha d_\nu^\beta \underbrace{\text{tr}(X^\alpha X^\beta)}_{(*)} = \delta_{\alpha\beta} d_\mu^\alpha d_\nu^\beta$$

(*) In the CCWZ construction, one typically normalises the generators in such a way that

$$\text{tr}(X^\alpha X^\beta) = \delta_{\alpha\beta}$$

$$\text{tr}(d_\mu d^\mu) = \delta_{\alpha\beta} d_\mu^\alpha d^{\beta,\mu}$$

This is the only invariant term with two derivatives that we can write if R_π is irreducible.

This is, therefore, the **leading term of the EFT** if we organise the latter as an expansion in derivatives.

ii) We can of course try to write additional invariants, but they will involve more than two derivatives.

Example: $\left[\text{tr}(d_\mu d^\mu)\right]^2 \rightsquigarrow$ invariant, but with 4 derivatives

Example: We can use E_μ^a , via covariant derivatives. We can define

$$\nabla_\mu d_\nu^\alpha \equiv \partial_\mu d_\nu^\alpha + i E_\mu^a (t_\pi^a)_{\alpha\beta} d_\nu^\beta$$

and construct invariants as $(\nabla_\mu d_\nu^\alpha)(\nabla^\mu d^{\alpha,\nu})$, but this term has again 4 derivatives.

Added explanation (not in the original notes) — slip flagged

The original writes $(\nabla_\mu d_\nu^\alpha)(\nabla^\mu d^{\alpha,\mu})$; the repeated μ cannot be right (it would leave ν uncontracted and μ appearing three times). It is a slip for ν . Corrected above.

We now expand the leading invariant in terms of the Goldstone fields. We use the explicit form of the vielbein.

$$\begin{aligned} \text{tr} (d_\mu d^\mu) &= \delta_{\alpha\beta} d_\mu^\alpha d^{\beta,\mu} = \delta_{\alpha\beta} \left[\frac{\partial_\mu \pi^\alpha}{f_\pi} - \frac{1}{2f_\pi^2} (\partial_\mu \pi^\rho) \pi^\gamma f_{\rho\gamma\alpha} + \dots \right] \\ &\quad \times \left[\frac{\partial^\mu \pi^\beta}{f_\pi} - \frac{1}{2f_\pi^2} (\partial^\mu \pi^\rho) \pi^\gamma f_{\rho\gamma\beta} + \dots \right] = \\ &= \frac{1}{f_\pi^2} (\partial_\mu \pi^\alpha) (\partial^\mu \pi^\alpha) + \mathcal{O} \left(\frac{(\partial\pi)^2 \pi^2}{f_\pi^4} \right) \end{aligned}$$

Added explanation (not in the original notes) — Why the cross terms do not survive at this order

Worth one line, because the notes jump straight to the answer. Multiplying the two brackets gives three kinds of term:

- $\frac{1}{f_\pi^2} (\partial_\mu \pi^\alpha) (\partial^\mu \pi^\alpha)$ — the leading piece;
- two cross terms $\propto \frac{1}{f_\pi^3} (\partial\pi)(\partial\pi)\pi f$, which carry *three* fields and are $\mathcal{O}(\pi/f_\pi)$ relative to the first;
- the square of the second brackets, $\mathcal{O}((\partial\pi)^2 \pi^2 / f_\pi^4)$.

The cross terms are actually *zero* by symmetry: $\delta_{\alpha\beta}$ forces the contraction $f_{\rho\gamma\alpha} (\partial_\mu \pi^\rho) \pi^\gamma (\partial^\mu \pi^\alpha)$, and $f_{\rho\gamma\alpha}$ is totally antisymmetric while $(\partial_\mu \pi^\rho) (\partial^\mu \pi^\alpha)$ is symmetric in $\rho \leftrightarrow \alpha$. So the first correction really is at $\mathcal{O}(\pi^2)$, as written. The $\mathcal{O}$ symbol here has dimension 2, matching the left-hand side: $[(\partial\pi)^2 \pi^2 / f_\pi^4] = (4 + 2) - 4 = 2$. ✓

2 The EFT Lagrangian, and its cutoff

We thus write the EFT

[p. 11*****]

$$\begin{aligned}\mathcal{L}_\pi &= \frac{f_\pi^2}{2} \operatorname{tr}(d_\mu d^\mu) = \\ &= \underbrace{\frac{1}{2}(\partial_\mu \pi^\alpha)(\partial^\mu \pi^\alpha)}_{(1)} + \underbrace{\mathcal{O}\left(\frac{(\partial\pi)^2 \pi^2}{f_\pi^2}\right)}_{(2)}\end{aligned}$$

(1) **Canonically normalised kinetic term** for the Goldstone fields.

(2) The interactions **always contain derivatives** (no potential!).

An **infinite set** of two-derivative interactions, **all fixed by the symmetry** and controlled by a **unique parameter** f_π .

Obs: The Lagrangian is **non-renormalisable**, and must be understood as an EFT.

The π - π scattering mediated by the interaction $\mathcal{O}((\partial\pi)^2 \pi^2 / f_\pi^2)$ will give an amplitude that grows with energy as

$$i\mathcal{M} = \mathcal{O}\left(\frac{E^2}{f_\pi^2}\right)$$

$\leadsto$ The EFT is valid at energies $E \ll f_\pi$, and has a cutoff scale identified by f_π .

Added explanation (not in the original notes) — Reading the normalisation, and the two things f_π is doing

Why the prefactor is $f_\pi^2/2$. From [p. 10*****], $\operatorname{tr}(d_\mu d^\mu) = f_\pi^{-2}(\partial\pi)^2 + \dots$. Multiplying by $f_\pi^2/2$ is the *only* choice that makes the kinetic term canonical, $\frac{1}{2}(\partial\pi)^2$. Note that this is not a free choice of the theorist: the overall coefficient of the unique invariant is one real number, and fixing the kinetic term to be canonical uses it up. Everything else — every vertex, to all orders in π — is then *predicted*. That is the meaning of the magenta remark.

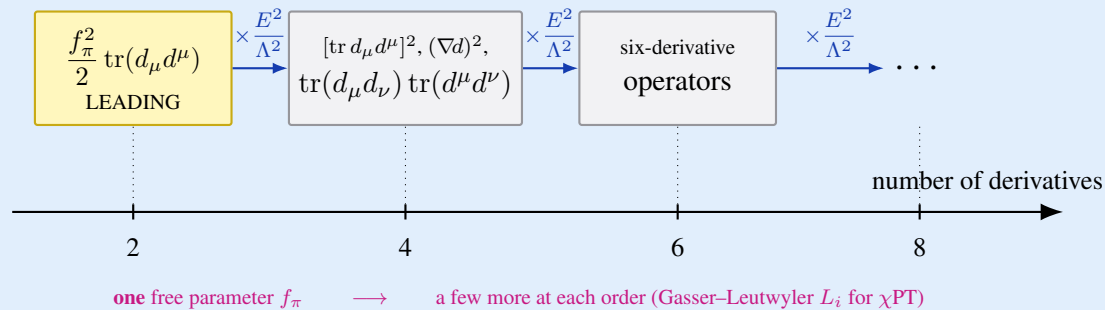
f_π **wears two hats**. It appeared first as a normalisation inside $U(\pi)$, purely to give π^α mass dimension 1. Here it reappears as the *physical cutoff*: the scale at which the derivative expansion, and with it the whole EFT, breaks down. These are the same number, and that is not a coincidence — there is only one scale in the problem.

The estimate $i\mathcal{M} \sim E^2/f_\pi^2$, in one line. A four-point vertex from $(\partial\pi)^2 \pi^2 / f_\pi^2$ has two derivatives and $1/f_\pi^2$, so at tree level each derivative brings down a momentum $\sim E$ and $\mathcal{M} \sim E^2/f_\pi^2$. Unitarity of the partial-wave expansion requires $|\mathcal{M}| \lesssim \mathcal{O}(1)$ (more precisely 16π), hence $E \lesssim \sqrt{16\pi} f_\pi$; the standard refinement of the naive $E \ll f_\pi$ is $\Lambda \simeq 4\pi f_\pi$. The notes state the estimate in the loose form; the 4π is naive-dimensional-analysis bookkeeping for the loop factor.

Sanity check against a known case. For QCD pions $f_\pi \simeq 92$ MeV, so $\Lambda \simeq 4\pi f_\pi \simeq 1.2$ GeV — and indeed chiral perturbation theory works beautifully below that and dies right around the ρ meson, $m_\rho = 775$ MeV. The Lagrangian above, with $G/H = SU(2)_L \times SU(2)_R / SU(2)_V$, is leading-order χ PT.

Added explanation (not in the original notes) — Figure — what “the derivative expansion” means, drawn

Each additional pair of derivatives costs a factor E^2/Λ^2 . The tower of terms is ordered, and the ordering is what makes the theory predictive at low energy.



The picture is only useful because the arrows point *downwards* in size. If $E \gtrsim \Lambda$ every box contributes equally, the ordering collapses, and there is no predictive theory left — which is precisely the statement “the EFT has a cutoff at f_π ”.

$\rightsquigarrow$ Terms with more than two derivatives. For example, a term $\mathcal{O}\left(\frac{(\partial\pi)^4}{f_\pi^4}\right)$ gives contributions that, with respect to the leading interactions, go like

$$\mathcal{O}\left(\frac{(\partial\pi)^2\pi^2}{f_\pi^2}\right) + \mathcal{O}\left(\frac{(\partial\pi)^4}{f_\pi^4}\right) \sim \frac{E^2}{f_\pi^2} \left(1 + \frac{E^2}{f_\pi^2}\right)$$

Since in the regime of validity of the EFT we have $E \ll f_\pi$, this is just a small correction compared to the leading term, and the derivative expansion makes sense.

Added explanation (not in the original notes) — Reading the schematic estimate

The chain is deliberately schematic — it is an estimate of *ratios*, not an equation. Read it as: an amplitude computed from the two-derivative Lagrangian scales as E^2/f_π^2 ; adding a four-derivative operator multiplies that by $(1 + E^2/f_\pi^2)$. So the four-derivative terms are not a competing leading effect, they are the *first correction*, suppressed by $E^2/f_\pi^2 \ll 1$.

This is the logical closure of the promise made in Lecture 28 — “we will comment at the end on the validity of this criterium”. The criterion (organise by number of derivatives) is justified *a posteriori*: it is a good ordering exactly in the regime where the theory is being used, and it fails exactly where the theory fails. The two statements are the same statement.

One subtlety the notes pass over. Loops of the two-derivative Lagrangian also generate four-derivative structures, with divergences absorbed into the four-derivative coefficients. This is why a non-renormalisable theory is still perfectly well defined order by order: at each order in E^2/Λ^2 only finitely many counterterms are needed. This is Weinberg’s power-counting theorem (1979), and it is what turned “non-renormalisable” from a death sentence into a bookkeeping scheme.

3 Final considerations [p. 12^{*****}][p. 13^{*****}]

~> FINAL CONSIDERATIONS

What we've learned in the past lectures:

Goldstone bosons are important because they are the universal low-energy footprints of spontaneously broken continuous symmetries

↪ When a physical system has a continuous symmetry in its laws but its ground state does not respect that symmetry, the system develops low-energy excitations: **Goldstone bosons**. [p. 13^{*****}]

This is a powerful idea because Goldstone bosons are controlled mostly by symmetry, not by microscopic details

↪ Many different systems can have the same kind of low-energy physics if they share the same symmetry-breaking pattern

That is where effective field theory becomes essential

↪ At low energies, we do not need the full microscopic theory: we write an EFT using only the relevant DOF (in this case the Goldstone bosons) and impose the symmetries.

Added explanation (not in the original notes) — Figure — universality, and why the same Lagrangian describes all of these

The claim “many different systems, same low-energy physics” is worth making concrete. Every row below has a completely different microscopic theory; all of them are described at low energy by the *same* Lagrangian $\frac{f_\pi^2}{2} \text{tr}(d_\mu d^\mu)$, differing only in $G \rightarrow H$ and in the number f_π .

system	breaking $G \rightarrow H$	# GBs	the Goldstones are
Ferromagnet	$SO(3) \rightarrow SO(2)$	$2 \rightarrow 1^\dagger$	magnons (spin waves)
Superfluid ^4He	$U(1) \rightarrow \mathbb{K}$	1	phonons
QCD, 2 flavours	$SU(2)_L \times SU(2)_R \rightarrow SU(2)_V$	3	the pions $\pi^\pm, \pi^0$
Crystal lattice	translations $\rightarrow$ discrete	3	acoustic phonons
BCS superconductor	$U(1) \rightarrow \mathbb{K}$	1	<i>eaten</i> — see Part II

[†] The ferromagnet is the standard cautionary case: two generators are broken but there is only *one* magnon (with quadratic dispersion), so the naive count $\dim G - \dim H$ fails — at finite charge density two broken generators can pair into a single “type-B” Goldstone. Not covered in these lectures — mentioned so the table is not misleading.

The last row is the bridge to the rest of this scan: in a superconductor the broken symmetry is *gauged*, and the would-be Goldstone does not appear as a massless particle at all. It is eaten, and the photon inside the superconductor becomes massive — the Meissner effect. That is the Higgs mechanism, discovered in condensed matter before particle physics.

Added explanation (not in the original notes) — The whole Goldstone arc of Lectures 24–29 in six lines

1. A continuous symmetry of $\mathcal{L}$ not shared by the vacuum $\Rightarrow$ massless scalars, one per broken generator (Goldstone's theorem, L.24–25).
2. The vacuum manifold is the coset $\mathcal{M}_0 \cong G/H$; the Goldstones are coordinates on it (L.26–27).
3. The Goldstone matrix $U(\pi)$ is a coset representative; G acts on it by $gU = U(\pi^g)h(\pi, g)$ (L.27–28).

4. $U^{-1}\partial_\mu U$ splits into a vielbein d_μ^α (transforms homogeneously) and a connection E_μ^a (transforms as a gauge field) — CCWZ (L.28).
5. The unique two-derivative invariant is $\text{tr}(d_\mu d^\mu)$; the leading EFT is $\frac{f_\pi^2}{2} \text{tr}(d_\mu d^\mu)$, valid up to $\Lambda \sim 4\pi f_\pi$ (**this lecture**).
6. Gauging the broken symmetry destroys the massless particles: they become longitudinal polarisations of massive gauge bosons (**the rest of this lecture**).

Added explanation (not in the original notes) — PDF page 6 is blank

The back of sheet [p. 13*****]. Nothing is missing.

PART II — The Higgs mechanism [p. 1*****]–[p. 8*****]

Added explanation (not in the original notes) — New chapter — the star count changes here

From this point on the handwritten page numbers restart at 1, so the margin markers carry *fourteen* stars, [p. 1*****]–[p. 8*****], rather than the thirteen of [p. 13*****] and everything before it. Nothing is missing; a new chapter simply begins.

Notation warning. In the previous chapter the Goldstone decay constant was f_π . In this chapter the same role is played by v , the vacuum expectation value, and the Goldstone is written $\pi(x)$ with π/v in the exponent. Do not let the reuse of the letter π confuse you: there is exactly *one* Goldstone here, because $G = U(1)$, $H = \mathbb{R}$, $\dim G - \dim H = 1$.

4 Setting up: gauge theory + spontaneous symmetry breaking [p. 1*****]

■ THE HIGGS MECHANISM

■ The abelian case

We combine a **gauge theory + spontaneous symmetry breaking**.

We consider the Lagrangian

$$\begin{aligned}\mathcal{L} &= (D_\mu \Phi)^\dagger (D^\mu \Phi) + \mu^2 \Phi^\dagger \Phi - \lambda (\Phi^\dagger \Phi)^2 - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} \\ D_\mu \Phi &= \partial_\mu \Phi + ig A_\mu \Phi \\ F_{\mu\nu} &= \partial_\mu A_\nu - \partial_\nu A_\mu \quad \mu^2 > 0, \quad \lambda > 0\end{aligned}$$

↔ **This theory is invariant under combined gauge transformations**

$$\begin{aligned}\Phi(x) &\longrightarrow \Phi'(x) = \exp(-i\alpha(x)) \Phi(x) \\ A_\mu(x) &\longrightarrow A'_\mu(x) = A_\mu(x) + \frac{1}{g} (\partial_\mu \alpha(x))\end{aligned}$$

↔ In addition, $\mu^2 > 0$ and $\lambda > 0$ **trigger spontaneous symmetry breaking**

All field configurations such that $\Phi^\dagger \Phi = \frac{\mu^2}{2\lambda} = \frac{v^2}{2}$ minimise the energy

[p. 1*****]

Added explanation (not in the original notes) — Check that the Lagrangian really is invariant, and the sign conventions

The covariant derivative. With $D_\mu\Phi = \partial_\mu\Phi + igA_\mu\Phi$ and $\Phi \rightarrow e^{-i\alpha}\Phi$:

$$\begin{aligned} D'_\mu\Phi' &= \partial_\mu(e^{-i\alpha}\Phi) + ig\left(A_\mu + \frac{1}{g}\partial_\mu\alpha\right)e^{-i\alpha}\Phi \\ &= e^{-i\alpha}\left[\underbrace{-i(\partial_\mu\alpha)\Phi + i(\partial_\mu\alpha)\Phi}_{=0} + \partial_\mu\Phi + igA_\mu\Phi\right] = e^{-i\alpha}D_\mu\Phi. \end{aligned}$$

So $D_\mu\Phi$ transforms exactly like Φ — that is the whole point of the word “covariant” — and $(D_\mu\Phi)^\dagger(D^\mu\Phi)$ is invariant. $F_{\mu\nu}$ is invariant because the added term $\frac{1}{g}\partial_\mu\partial_\nu\alpha$ is symmetric and F is antisymmetric.

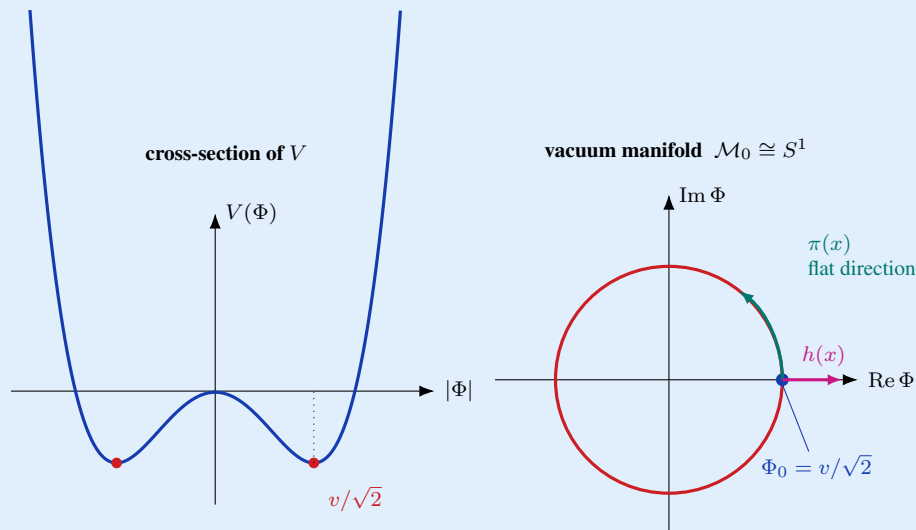
The sign of the mass term. Note carefully: the Lagrangian has $+\mu^2\Phi^\dagger\Phi$ with $\mu^2 > 0$, i.e. the potential is $V = -\mu^2\Phi^\dagger\Phi + \lambda(\Phi^\dagger\Phi)^2$, a *wrong-sign* mass. This is what makes $\Phi = 0$ a maximum rather than a minimum. Minimising:

$$\frac{\partial V}{\partial(\Phi^\dagger\Phi)} = -\mu^2 + 2\lambda\Phi^\dagger\Phi = 0 \implies \Phi^\dagger\Phi = \frac{\mu^2}{2\lambda} \equiv \frac{v^2}{2}, \quad v = \sqrt{\frac{\mu^2}{\lambda}}.$$

$\lambda > 0$ is needed for the potential to be bounded below.

What is genuinely new here. Nothing in the Lagrangian is new — it is the abelian Higgs model, i.e. the scalar QED of Lecture 22 with the mass term flipped. What is new is the *combination*: in Lectures 24–29 the broken symmetry was *global*, and the Goldstones were physical massless particles. Here it is *local*. The entire content of this chapter is that this one change — global $\rightarrow$ local — makes the Goldstone disappear and the photon heavy.

Added explanation (not in the original notes) — Figure — the potential, the vacuum circle, and what a gauge symmetry does to it



Two directions at the minimum: **radial** (curvature $\Rightarrow$ massive, the Higgs h) and **angular** (flat $\Rightarrow$ massless, the Goldstone π). That much is Lecture 24.

The new ingredient in one sentence. Because the symmetry is now *gauged*, the angle is not physical: two points on the red circle related by a phase rotation are not two different states, they are two *descriptions of the same state*. A gauge symmetry is a redundancy, not a symmetry. So the “flat direction” along which the Goldstone was supposed to be a physical excitation is a direction in which nothing physical happens at all — and one should not expect a physical massless particle. Making that precise is the content of [p. 2*****].

5 The crucial observation, and the unitary gauge [p. 2*****]

We make a choice of vacuum,

$$\langle \Phi \rangle = \Phi_0 = \frac{v}{\sqrt{2}}$$

[p. 2*****]

and introduce shifted fields in the exponential parametrisation:

$$\Phi(x) = \frac{1}{\sqrt{2}}(v + h(x)) e^{i\pi(x)/v}$$

$$\Phi(x) = \frac{1}{\sqrt{2}}(v + \underbrace{h(x)}_{\text{Higgs field}}) e^{i\frac{\pi(x)}{v}} \quad \underbrace{\hspace{1.5cm}}_{\text{Goldstone mode}}$$

We should now write the Lagrangian in the broken phase.

However, we make a **crucial observation**.

The theory is designed to be **gauge invariant**, meaning that the physics remains unchanged if we take, instead of $\Phi(x)$ and $A_\mu(x)$, the fields

$$\Phi'(x) = \underbrace{e^{-i\alpha(x)} e^{i\pi(x)/v}}_{(\star)} \frac{1}{\sqrt{2}} (v + h(x))$$

$$A'_\mu(x) = A_\mu(x) + \frac{1}{g} \partial_\mu \alpha$$

($\star$) Since this is true for all $\alpha(x)$, we make a **special choice**:

$$\alpha(x) = \frac{\pi(x)}{v} \quad \text{“unitary gauge”}$$

Added explanation (not in the original notes) — This is the pivot of the entire chapter — read it slowly

The move, stripped down. The Goldstone $\pi(x)$ enters Φ *only* as a phase, $e^{i\pi/v}$. A gauge transformation multiplies Φ by *another* phase, $e^{-i\alpha(x)}$, with $\alpha(x)$ completely arbitrary and x -dependent. So one may simply choose $\alpha = \pi/v$ and the two phases cancel exactly:

$$e^{-i\pi(x)/v} e^{i\pi(x)/v} = 1.$$

The Goldstone field is *gauged away*. Not approximately, not to some order in perturbation theory — identically, and for every field configuration.

Why this was impossible before. In Lectures 24–29 the symmetry was global: α had to be a *constant*. A constant α can shift $\pi(x)$ by a constant ($\pi \rightarrow \pi - v\alpha$, the non-linear shift symmetry), but it cannot remove an x -dependent field. That is exactly why the Goldstone was a physical propagating particle then. The single word “local” is the entire difference.

But degrees of freedom cannot vanish. This is the right thing to be suspicious about, and the notes are, on [p. 5*****]–[p. 6*****]. Nothing was destroyed — the count is conserved. What happened is that the same gauge freedom that removed π was previously being used to remove the longitudinal polarisation of A_μ . Spend it on π and you can no longer spend it on A_μ : the gauge field keeps its third polarisation, and a gauge field with three polarisations is a *massive* gauge field. The bookkeeping is on [p. 5*****].

Why the name “unitary”. Because in this gauge every field left in the Lagrangian is a physical particle — there are no unphysical states, so unitarity of the S-matrix is manifest. The price is that the propagator $\frac{-i}{k^2 - m_A^2} (g^{\mu\nu} - \frac{k^\mu k^\nu}{m_A^2})$ grows at large k , so renormalisability is *not* manifest. Other gauges (the R_ξ gauges) make the opposite trade: the Goldstone reappears as an explicit unphysical field, propagators behave well, but the physical spectrum is obscured. Both give the same S-matrix. This lecture uses unitary gauge throughout.

6 Substituting into the Lagrangian [p. 3*****]–[p. 4*****]

We are “fixing the gauge”.

If we make such choice, in the unitary gauge the fields are

[p. 3*****]

$$\begin{aligned}\Phi_u(x) &= \frac{1}{\sqrt{2}}(h + v) \\ A_u^\mu(x) &= A^\mu(x) + \frac{1}{v g} \partial^\mu \pi\end{aligned}$$

Note that with this gauge choice the Goldstone field “disappears” from the Lagrangian.

→ We now substitute in the Lagrangian

$$\begin{aligned}\mathcal{L} &= (\partial_\mu \Phi_u + ig A_{u,\mu} \Phi_u)^\dagger (\partial^\mu \Phi_u + ig A_u^\mu \Phi_u) \\ &\quad + \mu^2 \Phi_u^\dagger \Phi_u - \lambda (\Phi_u^\dagger \Phi_u)^2 - \frac{1}{4} F_{u,\mu\nu} F_u^{\mu\nu}\end{aligned}$$

► Since $\Phi \rightarrow \Phi_u$ and $A^\mu \rightarrow A_u^\mu$ is a gauge transformation, the Lagrangian takes the same form with fields now written in the unitary gauge.

We now substitute $\Phi_u(x) = \frac{1}{\sqrt{2}}(v + h(x))$ and look at the various pieces of the Lagrangian:

■ Potential terms

This is the usual computation, and gives the usual result

[p. 4*****]

$$V(h) = \frac{1}{2}(2\mu^2)h^2 + v\lambda h^3 + \frac{\lambda}{4}h^4$$

Added explanation (not in the original notes) — the potential computation, written out

The notes call it “usual” because it is identical to Lectures 24 and 27. For completeness, with $\Phi_u^\dagger \Phi_u = \frac{1}{2}(v + h)^2$:

$$\begin{aligned}V &= -\mu^2 \frac{1}{2}(v + h)^2 + \lambda \frac{1}{4}(v + h)^4 \\ &= -\frac{\mu^2}{2}(v^2 + 2vh + h^2) + \frac{\lambda}{4}(v^4 + 4v^3h + 6v^2h^2 + 4vh^3 + h^4).\end{aligned}$$

Using $\mu^2 = \lambda v^2$: the terms linear in h are $-\mu^2 v h + \lambda v^3 h = 0$ ✓ (as they must be — we expanded about a minimum), the constant is an irrelevant $-\frac{\lambda}{4}v^4$, and

$$h^2 : -\frac{\mu^2}{2} + \frac{3\lambda v^2}{2} = \lambda v^2 = \mu^2 \implies \frac{1}{2}(2\mu^2)h^2, \quad h^3 : \lambda v h^3, \quad h^4 : \frac{\lambda}{4}h^4.$$

So $m_h^2 = 2\mu^2 = 2\lambda v^2$ — the same Higgs mass as in the ungauged theory. The gauging does *not* touch the potential; it only touches the kinetic terms. That is why the next section is where all the action is.

■ Kinetic terms

Let us compute the term with the covariant derivative

$$D_\mu \Phi_u = \partial_\mu \Phi_u + ig A_{u,\mu} \Phi_u = \frac{1}{\sqrt{2}} \left[(\partial_\mu h) + ig A_{u,\mu} (h + v) \right]$$



$$\begin{aligned}
 (D_\mu \Phi_u)^\dagger (D^\mu \Phi_u) &= \frac{1}{2} \left[\partial_\mu h - ig A_{u,\mu} (h + v) \right] \left[\partial^\mu h + ig A_u^\mu (h + v) \right] = \\
 &= \frac{1}{2} (\partial_\mu h) (\partial^\mu h) + \frac{g^2}{2} A_{u,\mu} (h + v)^2 A_u^\mu = \\
 &= \frac{1}{2} (\partial_\mu h) (\partial^\mu h) + \frac{g^2 v^2}{2} A_{u,\mu} A_u^\mu + \frac{g^2}{2} (h^2 + 2vh) A_u^\mu A_{u,\mu}
 \end{aligned}$$

$$(D_\mu \Phi_u)^\dagger (D^\mu \Phi_u) = \frac{1}{2} (\partial_\mu h) (\partial^\mu h) + \frac{1}{2} g^2 v^2 A_{u,\mu} A_u^\mu + \frac{g^2}{2} (h^2 + 2vh) A_u^\mu A_{u,\mu}$$

We thus arrive at the following Lagrangian, written in the broken phase and in unitary gauge

Added explanation (not in the original notes) — Where the cross terms went — and the term that is *not* there

The cross terms cancel. Expanding the product, the two mixed terms are

$$+ \frac{ig}{2} (\partial_\mu h) A_u^\mu (h + v) - \frac{ig}{2} A_{u,\mu} (h + v) (\partial^\mu h) = 0,$$

identically, because they are complex conjugates of each other with opposite signs. This is the payoff of Φ_u being *real*: with a complex Φ these terms would survive.

The term that is conspicuously absent. In any other gauge one finds a term $\propto gv A^\mu \partial_\mu \pi$ — a *kinetic mixing* between the gauge field and the Goldstone, which is what makes the spectrum hard to read off (one has to diagonalise, and that is precisely what the R_ξ gauge-fixing term is designed to cancel). In unitary gauge it is simply not there, because π is not there. This is the practical reason the notes choose this gauge before substituting rather than after.

Reading the three surviving terms.

$\frac{1}{2} (\partial_\mu h) (\partial^\mu h)$	kinetic term for h	(canonically normalised ✓)
$\frac{1}{2} g^2 v^2 A_\mu A^\mu$	a mass term for the gauge field	the whole point
$\frac{g^2}{2} (h^2 + 2vh) A_\mu A^\mu$	hAA and $hhAA$ interactions	used on [p. 7*****]

The middle term came from the v^2 inside $(h + v)^2$ — i.e. *directly from the vev*. No vev, no gauge boson mass. Note also that all three come from the *same* single term $|D_\mu \Phi|^2$ with the *same* coupling g : the mass of the gauge boson and its couplings to the Higgs are rigidly locked together. This is a sharp, falsifiable prediction, and it is the reason measuring the hVV coupling at the LHC is a test of the mechanism and not just of the particle.

7 The broken-phase Lagrangian, and the counting of degrees of freedom [p. 5*****]–[p. 6*****]

$$\mathcal{L} = \underbrace{\frac{1}{2}(\partial_\mu h)(\partial^\mu h) - \frac{1}{2}(2\mu^2)h^2 - \lambda v h^3 - \frac{\lambda}{4}h^4}_{(1)} - \underbrace{\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}g^2v^2 A_{\mu}A^\mu}_{(2)} + \underbrace{\frac{1}{2}g^2(h^2 + 2hv) A_{\mu}A^\mu}_{(3)}$$

[p. 5*****]

(1) ► **Massive Higgs field with self-interactions**

(2) **Proca Lagrangian** describing a **massive spin-1 field** with mass

(3) **Interactions between the Higgs and the gauge field**

$$m_A^2 = g^2 v^2$$

↔ In the unitary gauge, there is no trace of the Goldstone bosons
However, the gauge field becomes massive, and it gains 1 DOF

↪ We say that the Goldstone boson “has been eaten” by the gauge field to become massive

The counting of DOF makes sense: we started with a complex scalar field and a massless gauge field (2 + 2 = 4 DOF) and we ended up with 1 real massive scalar field (1 DOF) + one massive gauge field (3 DOF).

[p. 6*****]

Added explanation (not in the original notes) — The mass, the factor of 2, and a warning about conventions

Why $m_A^2 = g^2 v^2$ and not $\frac{1}{2}g^2 v^2$. A Proca Lagrangian for a real vector is

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}m_A^2 A_\mu A^\mu.$$

Comparing with term (2), $\frac{1}{2}g^2 v^2 A_\mu A^\mu$, reads off $m_A^2 = g^2 v^2$ directly. The $\frac{1}{2}$ is part of the *definition* of the Proca mass term, not part of the mass. ✓

Conventions warning. With $\Phi = \frac{1}{\sqrt{2}}(v + h)e^{i\pi/v}$ one gets $m_A = gv$. Textbooks that write $\Phi \rightarrow \frac{v}{\sqrt{2}}$ with the covariant derivative $D_\mu = \partial_\mu - \frac{i}{2}gA_\mu$ (electroweak conventions, hypercharge $\frac{1}{2}$) get $m_A = \frac{1}{2}gv$. Both are right; the factor lives in the charge assignment. In the Standard Model, $m_W = \frac{1}{2}gv$ with $v \simeq 246$ GeV.

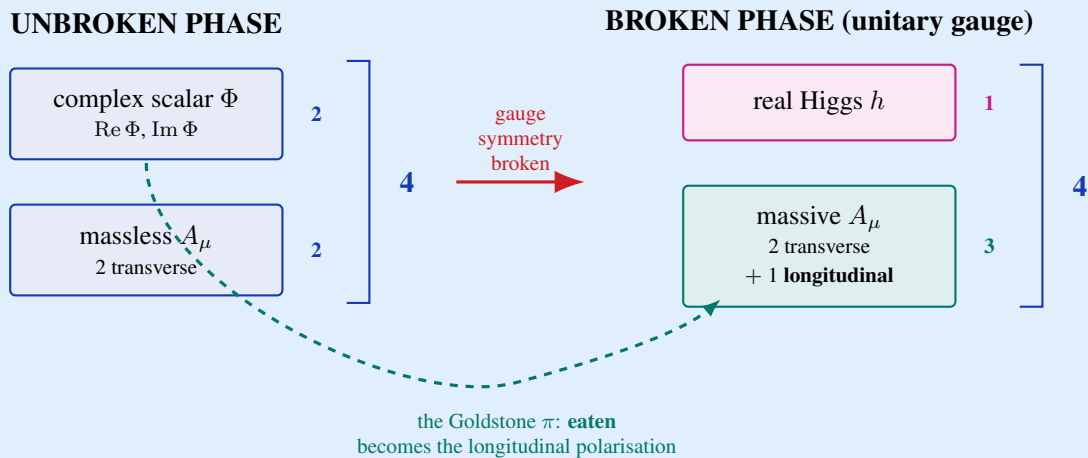
Everything is fixed by two numbers. The whole broken-phase spectrum is

$$m_h^2 = 2\mu^2 = 2\lambda v^2, \quad m_A^2 = g^2 v^2,$$

so the ratio $m_h/m_A = \sqrt{2\lambda}/g$ is a ratio of couplings. There is no free parameter hiding anywhere: three parameters upstairs (μ^2, λ, g) become three downstairs (m_h, m_A, g), and every interaction is then predicted.

Added explanation (not in the original notes) — Figure — the degree-of-freedom bookkeeping, drawn

This is the argument to hold on to. Degrees of freedom are conserved; only their labels change.



Why a massless vector has 2 and a massive one has 3. A 4-vector A_μ has four components; the equations of motion remove one, leaving three polarisations for a massive particle. For a *massless* particle the leftover gauge freedom $A_\mu \rightarrow A_\mu + \partial_\mu \alpha$ removes one more, leaving two — the two helicities of the photon. Once A_μ is massive that gauge freedom has been spent (on removing π), so nothing removes the third polarisation. *The Goldstone did not disappear; it changed address.*

8 Observation: finding the eaten Goldstone [p. 6^{*****}]

OBSERVATION

If it is true that the Goldstone boson has been eaten, then we should find it somewhere inside the belly of the gauge boson.

We discuss **two arguments**:

1) Consider the gauge transformation

$$A_\mu^\nu = A_\mu^\nu + \frac{1}{v g} (\partial_\mu \pi)$$

i) We started with a **massless** gauge field with **2 transverse DOF**

ii) In the **unitary gauge**, the gauge field is **massive**, with two transverse and **one longitudinal DOF**

iii) In this decomposition, therefore, the scalar field $\pi(x)$ appears indeed as **the extra DOF that makes A_μ massive**

iv) There is actually more. We know that in this decomposition the field $\pi(x)$ plays the role of the longitudinal polarisation in the ultra-relativistic limit

Added explanation (not in the original notes) — Why point iv) is the sharp statement, and where it comes from

The longitudinal polarisation vector. For a massive vector of mass m_A moving along $\hat{z}$ with energy E and momentum k ,

$$\varepsilon_L^\mu(k) = \frac{1}{m_A} (k, 0, 0, E) \xrightarrow{E \gg m_A} \frac{k^\mu}{m_A} + \mathcal{O}\left(\frac{m_A}{E}\right).$$

So at high energy $\varepsilon_L^\mu \simeq k^\mu/m_A$: the longitudinal polarisation vector becomes *proportional to the*

momentum. And k^μ is exactly what ∂^μ becomes in momentum space — which is why $\partial^\mu \pi / (vg)$ in the boxed gauge transformation is literally the longitudinal piece of A_u^μ .

The two features that make it grow. $\varepsilon_L \sim k/m_A$ carries a factor E/m_A that grows with energy. Every external longitudinal leg in an amplitude brings one such factor. This is why longitudinal gauge boson scattering is the diagnostic process: it is the place where the theory is most stressed, and therefore the place where the mechanism must most visibly work. The whole of [p. 7*****]–[p. 8*****] is that check.

The name of the theorem. The precise statement is the **Goldstone boson equivalence theorem**: at energies $E \gg m_A$, the amplitude for emitting a longitudinal gauge boson equals the amplitude for emitting the corresponding would-be Goldstone, up to $\mathcal{O}(m_A/E)$,

$$\mathcal{M}(A_L(k_1) \cdots) = \mathcal{M}(\pi(k_1) \cdots) + \mathcal{O}\left(\frac{m_A}{E}\right).$$

“Inside the belly of the gauge boson” is exactly this theorem, stated in the notes’ idiom. The exercise on the next two sheets verifies it by direct computation.

9 The check: longitudinal gauge boson scattering

[p. 7*****]-[p. 8*****]

2) The discussion at point 1) leads to the conclusion that the Goldstone boson “eaten” by the gauge boson appears as the longitudinal DOF in the ultrarelativistic limit

[p. 7*****]

↔ We check this argument with the following exercise.

We consider the interaction Lagrangian

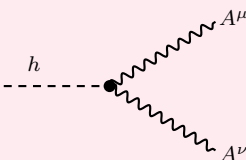
$$\mathcal{L}_{\text{int}} = \frac{g^2}{2} (h^2 + 2hv) A_\mu A^\mu$$

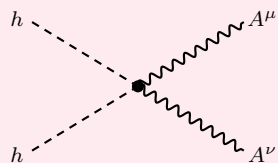
These are fields in the unitary gauge but we omit the “u” subscript.

and compute the longitudinal gauge boson scattering at the tree level.

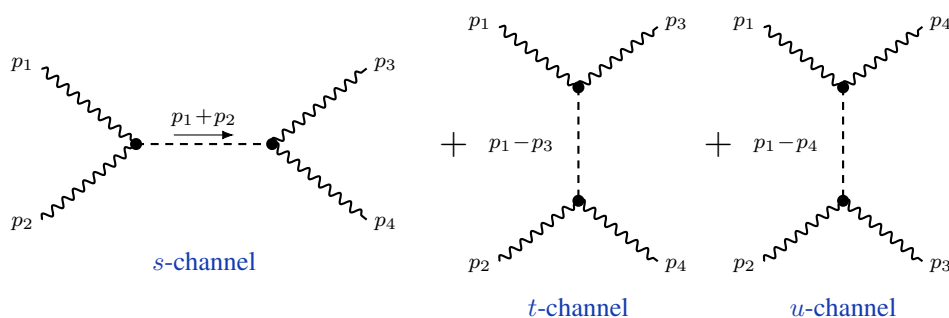
$$A_L(p_1) + A_L(p_2) \longrightarrow A_L(p_3) + A_L(p_4)$$

The Feynman rules are

$$\mathcal{L}_{\text{int}}^{(3)} = g^2 h v g_{\mu\nu} A^\mu A^\nu \implies \text{Feynman diagram} = 2i v g^2 g_{\mu\nu}$$


$$\mathcal{L}_{\text{int}}^{(4)} = \frac{g^2}{2} h^2 g_{\mu\nu} A^\mu A^\nu \implies \text{Feynman diagram} = 2i g^2 g_{\mu\nu}$$


The relevant Feynman diagrams are



[p. 8*****]

The amplitude for the s-channel diagram is

$$i\mathcal{M}_s = \left[(2ivg^2)^2 g_{\mu\nu} g_{\rho\sigma} \frac{i}{s - m_h^2} \right] \varepsilon_L^\mu(p_1) \varepsilon_L^\nu(p_2) \varepsilon_L^\rho(p_3)^* \varepsilon_L^\sigma(p_4)^*$$

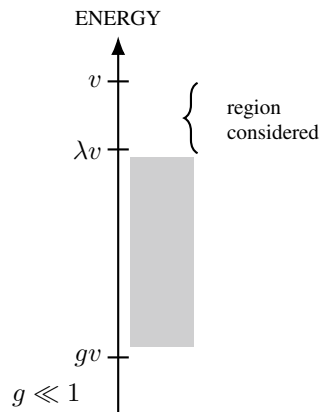
↓

$$i\mathcal{M}_s = (-i) 4v^2 g^4 \varepsilon_L^\mu(p_1) \varepsilon_L^\nu(p_2) \varepsilon_L^\rho(p_3)^* \varepsilon_L^\sigma(p_4)^* \left(\frac{g_{\mu\nu} g_{\rho\sigma}}{s - m_h^2} \right)$$

The full amplitude is

$$i\mathcal{M} = (-i) 4v^2 g^4 \varepsilon_L^\mu(p_1) \varepsilon_L^\nu(p_2) \varepsilon_L^\rho(p_3)^* \varepsilon_L^\sigma(p_4)^* \times \\ \left(\frac{g_{\mu\nu} g_{\rho\sigma}}{s - m_h^2} + \frac{g_{\mu\rho} g_{\nu\sigma}}{t - m_h^2} + \frac{g_{\mu\sigma} g_{\nu\rho}}{u - m_h^2} \right)$$

Suppose that we are in the following situation



We consider energies $s, t, u \gg m_A^2$ but **not necessarily smaller than the Higgs mass**

↪ **We use the UR limit for the gauge boson**

[the scan ends here]

10 Where this is going — completing the calculation

Added explanation (not in the original notes) — Everything in this section is added; it is not in the original notes

The scan breaks off exactly at the interesting moment. The setup is complete — the full amplitude is boxed on [p. 8*****] — and only the final substitution is missing. It is three lines, and the answer is the reason the exercise was set. Here it is, followed by the cross-check that verifies the claim of [p. 6*****], point iv).

Step 1 — put in the longitudinal polarisations

In the ultra-relativistic limit $E \gg m_A$ the longitudinal polarisation vector becomes proportional to the momentum,

$$\varepsilon_L^\mu(p) = \frac{p^\mu}{m_A} + \mathcal{O}\left(\frac{m_A}{E}\right), \quad m_A = gv.$$

Contract this with the three tensor structures, using $s = (p_1 + p_2)^2 = 2p_1 \cdot p_2$, $t = (p_1 - p_3)^2 = -2p_1 \cdot p_3$, $u = (p_1 - p_4)^2 = -2p_1 \cdot p_4$ (all masses neglected against E):

$$\varepsilon_L(p_1) \cdot \varepsilon_L(p_2) = \frac{p_1 \cdot p_2}{m_A^2} = \frac{s}{2g^2v^2}, \quad \varepsilon_L(p_1) \cdot \varepsilon_L(p_3)^* = \frac{p_1 \cdot p_3}{m_A^2} = \frac{-t}{2g^2v^2},$$

and similarly for the remaining pairs. Each channel therefore produces the square of one such factor, e.g. for the s -channel

$$\varepsilon_L^\mu(p_1) \varepsilon_L^\nu(p_2) g_{\mu\nu} \cdot \varepsilon_L^\rho(p_3)^* \varepsilon_L^\sigma(p_4)^* g_{\rho\sigma} = \frac{s}{2g^2v^2} \cdot \frac{s}{2g^2v^2} = \frac{s^2}{4g^4v^4}.$$

The prefactor $4v^2g^4$ cancels against $4g^4v^4$ up to $1/v^2$, and the result is remarkably clean:

$$i\mathcal{M} \xrightarrow{E \gg m_A} -\frac{i}{v^2} \left[\frac{s^2}{s - m_h^2} + \frac{t^2}{t - m_h^2} + \frac{u^2}{u - m_h^2} \right]$$

Note what has happened to the couplings: g has *completely disappeared*. The amplitude for scattering longitudinal *gauge bosons* depends only on v and m_h — i.e. only on the *scalar* sector. The gauge coupling cancelled between the vertex ($\propto g^2v$) and the two polarisation vectors ($\propto 1/m_A^2 = 1/g^2v^2$). This is the first sign that we are really scattering Goldstones.

Step 2 — the high-energy behaviour: the Higgs saves unitarity

Write each term as $\frac{x^2}{x - m_h^2} = x + m_h^2 + \frac{m_h^4}{x - m_h^2}$ and use the kinematic identity $s + t + u = \sum_i m_i^2 = 4m_A^2$:

$$i\mathcal{M} = -\frac{i}{v^2} \left[\underbrace{4m_A^2 + 3m_h^2}_{\text{constant}} + m_h^4 \left(\frac{1}{s - m_h^2} + \frac{1}{t - m_h^2} + \frac{1}{u - m_h^2} \right) \right] \xrightarrow{E \gg m_h} -i(4g^2 + 6\lambda).$$

Added explanation (not in the original notes) — This is the punchline — read the cancellation carefully

Each diagram separately grows. At large s , the s -channel alone gives $-is/v^2 \sim -iE^2/v^2$, which grows without bound and would violate unitarity at $E \sim \sqrt{16\pi}v$. The same for t and u .

The sum does not. Because $s + t + u$ is a *constant* fixed by the external masses, the three growing pieces cancel exactly and the total amplitude tends to a constant, $-i(4g^2 + 6\lambda)$ — pure couplings, no energy. Unitarity is safe at all energies.

Why this is a non-trivial statement about the mechanism. The cancellation works only because the *same* v appears in three places: in the gauge boson mass $m_A = gv$ (hence in $\varepsilon_L \simeq p/gv$), in the hAA vertex $2ivg^2$, and in the Higgs self-coupling. Change any one of them independently — give A_μ a Proca mass by hand and keep the scalar, or rescale the hAA coupling away from its predicted value — and the three E^2 pieces no longer cancel. *The Higgs exchange is not an optional extra; it is what keeps a massive gauge boson sensible at high energy.*

An honest caveat about this abelian toy. Here the *only* tree diagrams are the three Higgs exchanges: an abelian gauge field has no self-interactions. So if one removed the Higgs entirely ($m_h \rightarrow \infty$) the amplitude would go to zero rather than blow up — which is correct, since a single Goldstone on the flat coset $U(1) \cong S^1$ is a free field. The statement above is therefore about the *structure* of the cancellation, not about a disaster in this particular model.

Where the same structure does real work. In the non-abelian case — the Standard Model — the triple and quartic gauge vertices contribute to $W_L W_L \rightarrow W_L W_L$ and generate growth like E^4 and E^2 . The E^4 pieces cancel among the gauge diagrams themselves; the leftover E^2/v^2 is cancelled *only* by Higgs exchange, by exactly the mechanism shown above. Demanding that what remains respect partial-wave unitarity, $|a_0| \leq \frac{1}{2}$, bounds the Higgs mass:

$$m_h \lesssim \mathcal{O}(1 \text{ TeV}) \quad \text{for } v = 246 \text{ GeV},$$

the **Lee–Quigg–Thacker bound** (1977). This is the argument that guaranteed the LHC would find *something* below about a TeV. The Higgs must not only exist, it must be light — and it is, at 125 GeV.

Step 3 — the cross-check: scatter the Goldstones instead

This is the actual point of the exercise, and it is worth doing because it is short. Go back *before* gauge fixing and switch the gauge coupling off, $g \rightarrow 0$, so that π is an honest massless scalar. From $|\partial_\mu \Phi|^2$ with $\Phi = \frac{1}{\sqrt{2}}(v + h)e^{i\pi/v}$:

$$|\partial_\mu \Phi|^2 = \frac{1}{2}(\partial_\mu h)(\partial^\mu h) + \frac{1}{2}\left(1 + \frac{h}{v}\right)^2 (\partial_\mu \pi)(\partial^\mu \pi) \supset \frac{h}{v} (\partial_\mu \pi)(\partial^\mu \pi).$$

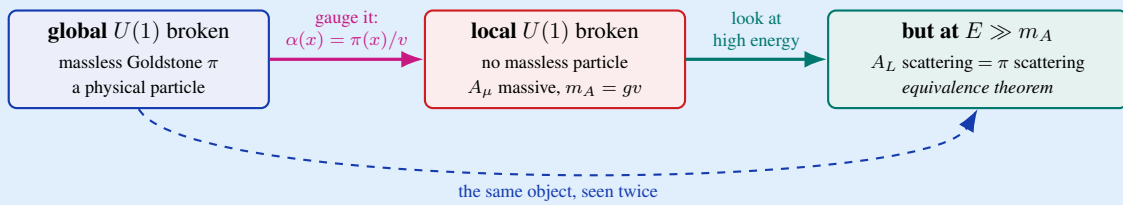
The $h\pi\pi$ vertex is then $-is/v$ for two incoming Goldstones of invariant mass s (each derivative brings a momentum), so the s -channel Higgs exchange gives

$$i\mathcal{M}_s^{\pi\pi} = \left(-\frac{is}{v}\right)^2 \frac{i}{s - m_h^2} = -\frac{i}{v^2} \frac{s^2}{s - m_h^2},$$

and likewise for t and u . Summing:

$$i\mathcal{M}(\pi\pi \rightarrow \pi\pi) = -\frac{i}{v^2} \left[\frac{s^2}{s - m_h^2} + \frac{t^2}{t - m_h^2} + \frac{u^2}{u - m_h^2} \right] = i\mathcal{M}(A_L A_L \rightarrow A_L A_L) \Big|_{E \gg m_A}$$

Term by term, identically. Scattering four longitudinal gauge bosons at high energy is the same calculation as scattering four Goldstone bosons. That is the **Goldstone boson equivalence theorem**, and it is the precise, quantitative form of the sentence on [p. 6^{*****}]: *the eaten Goldstone is found inside the belly of the gauge boson.*

Added explanation (not in the original notes) — Figure — the whole chapter in one picture

The Goldstone is never destroyed. It is relabelled — and at high energy, where the mass that did the relabelling is negligible, the original label reappears.

Summary of Lecture 29 — the logic in ten lines

Added explanation (not in the original notes) — Summary

Part I — closing CCWZ

1. $\text{tr}(d_\mu d_\nu)$ is invariant because $d_\mu \rightarrow h d_\mu h^{-1}$ and the trace is cyclic; with the normalisation $\text{tr}(X^\alpha X^\beta) = \delta_{\alpha\beta}$ it equals $\delta_{\alpha\beta} d_\mu^\alpha d^\beta{}_\mu$. If R_π is irreducible this is the *only* two-derivative invariant. [p. 9*****]
2. Everything else — $[\text{tr } d_\mu d^\mu]^2, (\nabla_\mu d_\nu^\alpha)(\nabla^\mu d^{\alpha,\nu})$ — carries four derivatives. Expanding the leading invariant gives $f_\pi^{-2}(\partial_\mu \pi^\alpha)(\partial^\mu \pi^\alpha) + \dots$. [p. 10*****]
3. Hence the EFT $\mathcal{L}_\pi = \frac{f_\pi^2}{2} \text{tr}(d_\mu d^\mu) = \frac{1}{2}(\partial\pi)^2 + \mathcal{O}((\partial\pi)^2 \pi^2 / f_\pi^2)$: a canonical kinetic term plus an *infinite* set of two-derivative interactions, all fixed by symmetry, all controlled by the single parameter f_π . No potential — every interaction has derivatives. [p. 11*****]
4. The theory is non-renormalisable; $\pi\pi$ amplitudes grow as E^2/f_π^2 , so the EFT is valid for $E \ll f_\pi$ with cutoff f_π (more precisely $\Lambda \sim 4\pi f_\pi$). Higher-derivative terms are suppressed by E^2/f_π^2 , so the derivative expansion is self-consistent exactly where it is used. [p. 11*****]–[p. 12*****]
5. *Final considerations.* Goldstone bosons are the universal low-energy footprints of spontaneously broken continuous symmetries, controlled by symmetry rather than microscopic detail — which is why systems as different as ferromagnets, superfluids and QCD share a low-energy description, and why EFT is the right language. [p. 12*****]–[p. 13*****]

Part II — the Higgs mechanism, abelian case

6. Take the abelian Higgs Lagrangian: scalar QED with $+\mu^2 \Phi^\dagger \Phi$, $\mu^2, \lambda > 0$. Gauge invariance under $\Phi \rightarrow e^{-i\alpha(x)}\Phi$, $A_\mu \rightarrow A_\mu + \frac{1}{g}\partial_\mu \alpha$; SSB with $\Phi^\dagger \Phi = v^2/2$. [p. 1*****]
7. Write $\Phi = \frac{1}{\sqrt{2}}(v + h)e^{i\pi/v}$. *Crucial observation:* since $\alpha(x)$ is arbitrary, choose $\alpha = \pi/v$ — the **unitary gauge** — and the Goldstone is removed from the Lagrangian identically. [p. 2*****]
8. Substituting $\Phi_u = \frac{1}{\sqrt{2}}(v + h)$: the potential gives the usual $V(h) = \frac{1}{2}(2\mu^2)h^2 + v\lambda h^3 + \frac{\lambda}{4}h^4$; the covariant derivative gives $\frac{1}{2}(\partial h)^2 + \frac{1}{2}g^2 v^2 A_\mu A^\mu + \frac{g^2}{2}(h^2 + 2vh)A_\mu A^\mu$. [p. 3*****]–[p. 4*****]
9. Reading off: a massive Higgs with self-interactions, a **Proca** term with $m_A^2 = g^2 v^2$, and $hAA/hhAA$ interactions. The Goldstone has vanished but the gauge field gained one DOF: it was **eaten**. Counting: $2 + 2 = 4$ before, $1 + 3 = 4$ after. [p. 5*****]–[p. 6*****]
10. Where is it? In the decomposition $A_u^\mu = A^\mu + \frac{1}{vg}\partial^\mu \pi$ the Goldstone is the longitudinal DOF; since $\varepsilon_L^\mu \rightarrow p^\mu/m_A$ at high energy, it *is* the longitudinal polarisation in the UR limit. The check — tree-level $A_L A_L \rightarrow A_L A_L$ via s, t, u Higgs exchange — confirms it: the amplitude equals the four-Goldstone amplitude term by term, and tends to a constant, so unitarity is preserved by the Higgs exchange. [p. 6*****]–[p. 8*****] (*last step completed in Section 10*)

Added explanation (not in the original notes) — Where this is going next

The abelian case is a warm-up with the full logical content but none of the group theory. The non-abelian generalisation is the obvious next step: $G \rightarrow H$ with $\dim G - \dim H$ would-be Goldstones, of which one is eaten for every *gauged* broken generator, leaving $\dim G - \dim H -$

(number of gauged broken generators) genuine massless Goldstones and that many massive gauge bosons. Applied to $SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{em}}$: three broken generators, all gauged, three eaten Goldstones, three massive vectors $W^\pm, Z$, one surviving massless photon, and one physical scalar left over — the Higgs boson. That is the electroweak theory, and every ingredient needed to state it has now been assembled over Lectures 24–29.

The other thread left open is the case where a symmetry is *both* spontaneously broken and gauged only partially — as in QCD, where the chiral symmetry is global and the pions survive as physical (pseudo-)Goldstones, but the electromagnetic subgroup is gauged, which is what splits $m_{\pi^\pm}$ from m_{π^0} .